

1) A manufacturing process claims that the average diameter of the ball bearing is $\mu_0 = 12 \text{ mm}$. A sample of $n = 40$ bearings has

- Sample mean: $\bar{x} = 11.7 \text{ mm}$
- Population standard deviation: $\sigma = 0.8$

At 5% significance, test whether the true mean diameter differs from 12 mm.

Solⁿ

Given:

Claimed mean $\mu_0 = 12 \text{ mm}$

Sample size $n = 40$

Sample mean $\bar{x} = 11.7 \text{ mm}$

Population std dev $\sigma = 0.8 \text{ mm}$

Significance level $\alpha = 0.05$

Here,

- Population std deviation (σ) is known.

- Sample size $n \geq 30$

- We are testing single population mean.

Hence "One sample z-test" is appropriate.

Hypothesis

$H_0 : \mu = 12 \text{ mm}$

$H_1 : \mu \neq 12 \text{ mm}$ (two tailed)

Hence using the formula

$$Z = \frac{\bar{x} - \mu_0}{\text{Std. Error}} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

$$= \frac{11.7 - 12}{0.8/\sqrt{40}}$$

$$\boxed{Z = -2.37}$$

For a two tailed z-test
Critical value at $\alpha = 0.05$

$$Z\text{-Critical} = \pm 1.96$$

Since $|Z| = 2.37 > 1.96$
we reject H_0 .

Conclusion: There is sufficient evidence that at 5% significance level that the true mean differs from 12 mm.

2) A hospital reports that the success rate of a surgery is 90%. A new sample of $n = 120$ patients shows that 98 recover successfully. Test at $\alpha = 0.05$ whether the success rate has changed.

Solⁿ

Given:

- Claimed success rate $p = 0.90$
 - sample size $n = 120$
 - Sample successes $x = 98$
- $$\hat{p} = \frac{98}{120} = 0.817$$

$$\alpha = 0.05$$

Here,

- Sample size is large ($n = 120$)
- It is a proportion test

Hypothesis

$$H_0: p = 0.90$$

$$H_1: p \neq 0.90$$

Test statistics

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

$\hat{p}$ - sample proportion

p - population proportion

n - no of samples.

$$\begin{aligned} \text{sd} &= \sqrt{\frac{p(1-p)}{n}} \\ &= \sqrt{\frac{(0.9)(0.1)}{120}} = 0.0274 \end{aligned}$$

$$Z = \frac{0.817 - 0.9}{0.0274}$$

$$Z = -3.03$$

At significance level $\alpha = 0.05$

$$Z_{\text{critical}} = \pm 1.96$$

since $|Z| = 3.03 > 1.96$
we reject the H_0 .

Conclusion:

There is sufficient evidence that the success rate has changed significantly from 90%.

3) Two drugs are tested for reduction of blood pressure.

• Drug A : $n_1 = 50$, $\bar{x}_1 = 18.4$, $\sigma_1 = 4.2$

• Drug B : $n_2 = 60$, $\bar{x}_2 = 16.9$, $\sigma_2 = 3.9$

Test at $\alpha = 0.01$ whether Drug A has a significantly different mean reduction compared to Drug B.

Solⁿ

Given

• Drug A $n_1 = 50$ $\bar{x}_1 = 18.4$ $\sigma_1 = 4.2$

• Drug B $n_2 = 60$ $\bar{x}_2 = 16.9$ $\sigma_2 = 3.9$

$\alpha = 0.01$

Here these are

- 2 independent samples
- Large sample size ($n_1, n_2 \geq 30$)
- Population std dev are known.

Hence we choose "two Samples z-test"

Hypothesis

$H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 \neq \mu_2$

for $\alpha = 0.01$ (two tailed)

the critical value

$z_{\text{critical}} = \pm 2.58$

Test

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad \swarrow \text{std. error}$$

$$= \frac{18.4 - 16.9}{\sqrt{\frac{(4.2)^2}{50} + \frac{(3.9)^2}{60}}}$$

$$= \frac{1.5}{\sqrt{0.353 + 0.254}}$$

$$= \frac{1.5}{0.779} = 1.93$$

Since $|z| = 1.96 < 2.58$

We fail to reject H_0

Conclusion: There is no significant difference between Drug A and Drug B at 1% Level.

4) Student take a test before and after a training program.

Student	Before	After
1	68	75
2	70	78
3	65	70
4	72	74
5	69	73

Test at $\alpha = 0.05$ whether the training improved the score.

Soln

Given

- Same students score before and after training
- Sample size $n = 5$ (small)
- Population standard deviation unknown.

Hypothesis.

$$H_0: \mu_d = 0$$

$$H_1: \mu_d > 0$$

Hence we move ahead with selecting "Paired T-test"

Create a difference table.

Student	Difference marks.
1	7
2	8
3	5
4	2
5	4

$$\bar{d} = \frac{\text{Mean difference}}{n} = \frac{(7+8+5+2+4)}{5} = 5.2$$

$$Sd = \sqrt{\frac{\sum (d_i - \bar{d})^2}{n-1}}$$

$$= \sqrt{\frac{(1.8^2 + 2.8^2 + 0.2^2 + 3.2^2 + 1.2^2)}{4}}$$

$$Sd = \sqrt{\frac{22.8}{4}} = 2.39$$

$$\text{Standard error} = \frac{Sd}{\sqrt{n}} = \frac{2.39}{\sqrt{5}} = 1.069$$

t-statistic

$$t = \frac{\bar{d}}{\text{std. error}} = \frac{5.2}{1.069} = 4.86$$

$$|t| = 4.86$$

$$\text{Degree of freedom} = n - 1 = 4$$

$$\text{At } \alpha = 0.05, \text{ and } df = 4$$

$$t_{\text{critical}} = 2.132$$

Since $t > t_{\text{critical}}$ ($4.86 > 2.132$) we Reject H_0 .

Conclusion: The training program significantly improved the test scores.

5) For the following scenario:

- Group 1 ($n=15$), $\bar{x}_1 = 52$, $s_1 = 8$
- Group 2 ($n=12$) $\bar{x}_2 = 46$, $s_2 = 5$

Test whether the means differ at $\alpha = 0.05$

Soln

Given

Group 1: $n_1 = 15$ $\bar{x}_1 = 52$ $s_1 = 8$

Group 2: $n_2 = 12$ $\bar{x}_2 = 46$ $s_2 = 5$

Here,

- Two groups are independent
- Sample size is smaller.
- Population std. dev are unknown.

Hence we choose "two sample t-test"
(unequal variances)

Hypotheses

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

test statistics.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$= \frac{52 - 46}{\sqrt{\frac{8^2}{15} + \frac{5^2}{12}}}$$

$$= \frac{6}{\sqrt{4.26 + 2.08}}$$

$$= \frac{6}{\sqrt{6.34}}$$

$$t = \frac{6}{2.51} = 2.39$$

$$\boxed{t = 2.39}$$

$$\begin{aligned} \text{Degree of freedom} &= n_1 + n_2 - 2 \\ &= 15 + 12 - 2 = 25 \end{aligned}$$

$$\text{At } \alpha = 0.05 \text{ and } df = 25$$
$$\boxed{t\text{-critical} = 2.060}$$

Since $|t| = 2.39 > 2.060$
we Reject H_0 .

Conclusion: There sufficient evidence at 5% significance level that the means of the two groups differ significantly.